AN INVERTIBLE MODULE WITH TRIVIAL GENERIC FIBRE OVER A FLAT FAMILY WITH INTEGRAL FIBRES ON A NORMAL BASE IS PULLED BACK FROM THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-in-image-pullback is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 6967-6981 and proof span 6983-7090 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof ends with the author's parenthesis '(some details omitted)' in the glueing paragraph; that phrase is kept verbatim and what it leaves to the reader is the bookkeeping of the closedness argument, while the affine computation and the construction of the module are carried out in the counted span.

Editorial note. Two of the cross-references in this proof point at the lane's own card S13-morph-fpqc-quotient-topology (the quotient topology statement for quasi-compact surjective flat morphisms) and at the openness statement for flat morphisms; both are delivered as navigation pointers.

Editorial note. 11 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A flat morphism with integral fibres over an integral base has integral source, cited from the same chapter. Cited at proof lines 6986-6987. Makes the trivializing section of the generic fibre into a regular meromorphic section on the source.
- Pullbacks of meromorphic functions along a morphism with integral fibres are defined, cited from the same chapter. Cited at proof lines 6988-6990. Allows the product of a pulled back meromorphic function with the chosen section to be considered.
- Structural facts about a normal local domain of dimension one: it is a discrete valuation ring, the intersection over height one primes with prescribed exponents equals a symbolic power expression, symbolic powers commute with flat base change and flat extensions intersect ideals, and a product of ideals equal to a principal ideal makes the factors finite locally free of rank

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one. Cited at proof lines 7011-7041. These are the algebra inputs of the affine computation that splits the section s into a product of a unit and an element coming from the base.
- The quotient topology statement for quasi-compact surjective flat morphisms and the openness statement for flat morphisms locally of finite presentation, cited from the morphisms chapter. Cited at proof lines 7066-7070. Decides the two cases of the hypothesis that the given morphism is either quasi-compact or locally of finite type, showing that the exceptional set is closed.

Source reference. Original lemma 00PD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 927–927. Same-chapter reference used in the retained author text at source line(s) 1819, 2358, 3021, 7023. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 031S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3995–3995. Same-chapter reference used in the retained author text at source line(s) 6899, 7019. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BBY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14261–14261. Same-chapter reference used in the retained author text at source line(s) 7046. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00KV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 765–765. Same-chapter reference used in the retained author text at source line(s) 2994, 3212, 7018. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 031T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3996–3996. Same-chapter reference used in the retained author text at source line(s) 3417, 6903, 6948, 7032. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 09ME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12177–12177. Same-chapter reference used in the retained author text at source line(s) 2541, 7053. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BC0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14263–14263. Same-chapter reference used in the retained author text at source line(s) 7045. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's

own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BCM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14285–14285. Same-chapter reference used in the retained author text at source line(s) 6989. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02OU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3529–3529. Same-chapter reference used in the retained author text at source line(s) 6096, 6992. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01UA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2403–2403. Same-chapter reference used in the retained author text at source line(s) 7088. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02JY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3353–3353. Same-chapter reference used in the retained author text at source line(s) 7086. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\varphi : X \to Y$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume that*

1. *X is locally Noetherian,*
2. *Y is locally Noetherian, integral, and normal,*
3. *φ is flat with integral (hence nonempty) fibres,*
4. *φ is either quasi-compact or locally of finite type,*
5. *$\mathcal{L}$ is trivial when restricted to the generic fibre of φ.*

Then $\mathcal{L} \cong \varphi^*\mathcal{N}$ for some invertible $\mathcal{O}_Y$-module $\mathcal{N}$.

Proof. Let $\xi \in Y$ be the generic point. Let X_ξ be the scheme theoretic fibre of φ over ξ. Denote $\mathcal{L}_\xi$ the pullback of $\mathcal{L}$ to X_ξ. Assumption (5) means that $\mathcal{L}_\xi$ is trivial. Choose a trivializing section $s \in \Gamma(X_\xi, \mathcal{L}_\xi)$. Observe that X is integral by Lemma 0BCM. Hence we can think of s as a regular meromorphic section of $\mathcal{L}$. Pullbacks of meromorphic functions are defined for φ by Lemma 02OU. Let $\mathcal{N} \subset \mathcal{K}_Y$ be the $\mathcal{O}_Y$-module whose sections over an open $V \subset Y$ are those meromorphic functions $g \in \mathcal{K}_Y(V)$ such that $\varphi^*(g)s \in \mathcal{L}(\varphi^{-1}V)$. A priori $\varphi^*(g)s$ is a section of $\mathcal{K}_X(\mathcal{L})$ over $\varphi^{-1}V$. We claim that $\mathcal{N}$ is an invertible $\mathcal{O}_Y$-module and that the map

$$\varphi^*\mathcal{N} \longrightarrow \mathcal{L}, \quad g \longmapsto gs$$

is an isomorphism.

We first prove the claim in the following situation: X and Y are affine and $\mathcal{L}$ trivial. Say $Y = \mathrm{Spec}(R)$, $X = \mathrm{Spec}(A)$ and s given by the element $s \in A \otimes_R K$ where K is the fraction field of R. We can write $s = a/r$ for some nonzero $r \in R$ and $a \in A$. Since s generates $\mathcal{L}$ on the generic fibre we see that there exists an $s' \in A \otimes_R K$ such

that $ss' = 1$. Thus we see that $s = r'/a'$ for some nonzero $r' \in R$ and $a' \in A$. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n \subset R$ be the minimal primes over rr'. Each $R_{\mathfrak{p}_i}$ is a discrete valuation ring (Algebra, Lemmas 00KV and 031S). By assumption $\mathfrak{q}_i = \mathfrak{p}_i A$ is a prime. Hence $\mathfrak{q}_i A_{\mathfrak{q}_i}$ is generated by a single element and we find that $A_{\mathfrak{q}_i}$ is a discrete valuation ring as well (Algebra, Lemma 00PD). Of course $R_{\mathfrak{p}_i} \to A_{\mathfrak{q}_i}$ has ramification index 1. Let $e_i, e_i' \geq 0$ be the valuation of a, a' in $A_{\mathfrak{q}_i}$. Then $e_i + e_i'$ is the valuation of rr' in $R_{\mathfrak{p}_i}$. Note that

$$\mathfrak{p}_1^{(e_1+e_1')} \cap \ldots \cap \mathfrak{p}_i^{(e_n+e_n')} = (rr')$$

in R by Algebra, Lemma 031T. Set

$$I = \mathfrak{p}_1^{(e_1)} \cap \ldots \cap \mathfrak{p}_i^{(e_n)} \quad \text{and} \quad I' = \mathfrak{p}_1^{(e_1')} \cap \ldots \cap \mathfrak{p}_i^{(e_n')}$$

so that $II' \subset (rr')$. Observe that

$$IA = (\mathfrak{p}_1^{(e_1)} \cap \ldots \cap \mathfrak{p}_i^{(e_n)})A = (\mathfrak{p}_1 A)^{(e_1)} \cap \ldots \cap (\mathfrak{p}_i A)^{(e_n)}$$

by Algebra, Lemmas 0BC0 and 0BBY. Similarly for $I'A$. Hence $a \in IA$ and $a' \in I'A$. We conclude that $IA \otimes_A I'A \to rr'A$ is surjective. By faithful flatness of $R \to A$ we find that $I \otimes_R I' \to (rr')$ is surjective as well. It follows that $II' = (rr')$ and I and I' are finite locally free of rank 1, see Algebra, Lemma 09ME. Thus Zariski locally on R we can write $I = (g)$ and $I' = (g')$ with $gg' = rr'$. Then $a = ug$ and $a' = u'g'$ for some $u, u' \in A$. We conclude that u, u' are units. Thus Zariski locally on R we have $s = ug/r$ and the claim follows in this case.

Let $y \in Y$ be a point. Pick $x \in X$ mapping to y. We may apply the result of the previous paragraph to $\mathrm{Spec}(\mathcal{O}_{X,x}) \to \mathrm{Spec}(\mathcal{O}_{Y,y})$. We conclude there exists an element $g \in R(Y)^*$ well defined up to multiplication by an element of $\mathcal{O}_{Y,y}^*$ such that $\varphi^*(g)s$ generates $\mathcal{L}_x$. Hence $\varphi^*(g)s$ generates $\mathcal{L}$ in a neighbourhood U of x. Suppose x' is a second point lying over y and $g' \in R(Y)^*$ is such that $\varphi^*(g')s$ generates $\mathcal{L}$ in an open neighbourhood U' of x'. Then we can choose a point x'' in $U \cap U' \cap \varphi^{-1}(\{y\})$ because the fibre is irreducible. By the uniqueness for the ring map $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x''}$ we find that g and g' differ (multiplicatively) by an element in $\mathcal{O}_{Y,y}^*$. Hence we see that $\varphi^*(g)s$ is a generator for $\mathcal{L}$ on an open neighbourhood of $\varphi^{-1}(y)$. Let $Z \subset X$ be the set of points $z \in X$ such that $\varphi^*(g)s$ does not generate $\mathcal{L}_z$. The arguments above show that Z is closed and that $Z = \varphi^{-1}(T)$ for some subset $T \subset Y$ with $y \notin T$. If we can show that T is closed, then g will be a generator for $\mathcal{N}$ as an $\mathcal{O}_Y$-module in the open neighbourhood $Y \setminus T$ of y thereby finishing the proof (some details omitted).

If φ is quasi-compact, then T is closed by Morphisms, Lemma 02JY. If φ is locally of finite type, then φ is open by Morphisms, Lemma 01UA. Then $Y \setminus T$ is open as the image of the open $X \setminus Z$. $\square$